Copy-Move Forgery Detection using a Siamese Swap-Test Quantum Neural Network

Phase-2 Progress & Evaluation Report

Final Year Project · Phase 2 | Date: 26 July 2026 | Pipeline: `cmfd_swaptest_pipeline.py` (v3 — pure QNN)

1. Executive Summary

This report documents the current state of a quantum-machine-learning pipeline for **copy-move forgery detection (CMFD)** — detecting regions of an image that have been duplicated and pasted elsewhere in the same image. The system pairs a classical patch descriptor with a **Siamese swap-test quantum neural network (QNN)** that learns a similarity score between two image patches, followed by a spatial offset-consistency filter that keeps only genuine duplications.

Headline finding. The pipeline is a genuine end-to-end quantum model, and it does not work as a detector. Both halves of that sentence are measured, not asserted:

- The trained circuit scores **AUC 0.883** against **0.887** for a parameter-free classical cosine similarity — **no quantum advantage**, at ~36 ms/pair versus microseconds.
- Training moves AUC by **0.003** (0.880 untrained → 0.883 trained). The 58 rotation angles influence the output *less* than the choice of input image does.
- Best achievable detection is **F1 0.0813** against a **0.0771** baseline of marking every pixel forged — and that is with an oracle per-image threshold.

Root cause. The classical offset-consistency vote removed in v3 was not scaffolding around the quantum model — **it was the component performing the detection**. Patch similarity alone drowns in natural self-similarity; a duplicated region is a small signal among tens of thousands of genuinely similar patch pairs. See §7.

v3 finding — the reported non-convergence was a data bug, not a model bug. Training loss was previously stuck at ~0.33, above the 0.25 constant-predictor floor, and this was attributed to the model. The actual cause was in synthetic pair construction: positives were rotated 90/180/270° three times out of four. A swap test compares amplitude vectors, so a rot90 duplicate permutes the state into a near-orthogonal one — **measured: an exact rot90 copy scores ~0.01 where an exact copy scores ~1.0**. Labelling those pairs 1.0 asks the circuit to fit a target its translation-only architecture provably cannot represent. With rotations excluded, **loss drops to ~0.003–0.10**, far below the floor. Rotated positives remain available behind `--train-rotations` and should only be re-enabled together with a dθ component in the offset register.

Status of §6–§10. The results, gap analysis, recommendations and roadmap below were produced with the **v2 hybrid** and have **not yet been re-run against v3**. They are retained as the

> v2 baseline for comparison. Treat every number in them as v2 until the v3 evaluation is complete; v3 is expected to score *lower*, since removing the offset vote removed the stage that actually suppressed false matches.

2. Objective & Target Output

The intended end product: *given any input image, decide whether it is forged, and if so, localize the exact duplicated region as a bounding box or heatmap.* That decomposes into four capabilities, tracked in the gap analysis (§8).

3. System Architecture (v3 — pure QNN)

The architecture is a **pairwise similarity model**, which is the structurally correct choice for copy-move (the task is inherently "does patch A match patch B elsewhere?"). A single-patch classifier cannot express duplication.

v3 removes every classical detection stage from the v2 hybrid. v2 was quantum only in its similarity readout: a classical DCT descriptor decided what the circuit was allowed to see, a classical sort decided which pairs it was allowed to score, and a classical Hough vote decided which of its outputs to believe. In v3 raw pixels enter the circuit and a measured expectation value is the decision.

Stage	v2 (hybrid)	v3 (pure)
Patch representation	24-dim DCT coefficients + intensity histogram	**AmplitudeEmbedding of the raw pixels** — all patch^2 values, L2-normalized state preparation only
Candidate pairing	O(N log N) lexicographic block matching (Fridrich)	**Exhaustive** — every admissible pair scored by the circuit, O(N^2)
False-match suppression	Classical (dy, dx) Hough vote	**In-circuit offset register** + trainable CRY coupling inside the interferometer
Mask cleanup	Morphological close (dilate→erode)	**None**
Trainable parameters	circuit rotation angles only	circuit rotation angles only (*unchanged — v2 was already pure here*)

Component	v3 specification
Quantum registers	**6 qubits per register** (8×8 patch = 64 amplitudes), two registers = 12 qubits **+ 2 offset qubits + 1 ancilla** = **15 wires total**
Trainable parameters	**58** — (3 layers × 6 qubits × 3 rotations) tied across both registers, plus (2 × 2) offset-register angles
Similarity readout	Swap test with an offset-conditioned readout: H → CSWAP → trainable CRY from the offset wires → H, measured as ancilla $\langle Z \rangle$, clipped to [0,1]
Patches	**8×8** px, stride **16** → 256 patches / ~33k pairs on a 256×256 image
Simulation	`lightning.qubit` + adjoint gradients (3–7× faster than `default.qubit`; measured **0.036 s per pair**)
Training	Self-supervised synthetic copy-move pairs, MSE on similarity labels

> **Do not "simplify" the ansatz.** A plain amplitude embedding followed by a data-independent tied unitary is provably degenerate under a swap test: $|\langle U(w)f_A | U(w)f_B \rangle|^2 = |\langle f_A | f_B \rangle|^2$ for *any* $U(w)$, so the gradient w.r.t. w is exactly zero. This was measured, not assumed: gradient norm **~1e-16** in the degenerate form vs **~1.2e-2** with data re-uploading. Re-uploading the pixels into the rotation angles is what keeps the tied weights alive.

Cost of purity. Removing the classical prefilter makes pair scoring $O(N^2)$, so stride is now the dominant cost dial: ~1 min/image at stride 32, ~20 min at stride 16, ~5 h at stride 8. The quantum replacement for the prefilter is Grover/amplitude amplification over a pair-index register, which requires QRAM to hold all patches in superposition — not simulable at this scale, so it is recorded as future work rather than substituted with a classical stand-in.

Known weakness, stated plainly. The offset register is a *weaker* mechanism than the vote it replaced, not an equivalent one. Offset consistency is a **global** property of the pair ensemble ("many pairs share one offset"); a per-pair circuit cannot observe clustering it is never shown. A faithful quantum version needs quantum counting over all pairs at once. Expect more false positives on homogeneous regions (sky, walls) than v2 produced.

4. Datasets & Their Roles

Training and evaluation must use **different** data sources — a recurring point of confusion. The Siamese similarity *must* be trained on synthetic pairs because real CMFD datasets provide no patch-level correspondence ground truth.

Dataset	Role	Why
Synthetic (self-supervised)	**Training — mandatory**	Only source of labeled positive/negative patch pairs; CoMoFoD masks have no correspondence labels
CoMoFoD (small v2)	**Primary evaluation (headline)**	Real forgeries + real binary masks; loader already matches it (5,000 forged images)
COVERAGE (100 pairs)	**Stress test only**	Adversarial by design (similar genuine objects); useful to prove generalization, but not a headline metric

Recommendation: train on synthetic, report on **CoMoFoD**. Never evaluate on synthetic data (it only proves the model learned its own augmentation). Keep COVERAGE as a "hard case" data point.

5. Evaluation Methodology — why not accuracy?

Forged regions occupy only **~2–11% of pixels**, so the pixel classes are extremely imbalanced. Under this imbalance, plain accuracy is **actively misleading**: a model that predicts "nothing is forged" scores ~90–98% accuracy while detecting **zero** forgeries.

Metric	Definition	Meaning here
Accuracy	(TP+TN) / all	**Misleading — rewards predicting nothing**
Recall	TP / (TP+FN)	Fraction of true forged pixels *found*. **Gamed by flagging everything** → this is the "80%" trap
Precision	TP / (TP+FP)	Of the pixels flagged, how many were actually forged
F1	2·P·R / (P+R)	**Honest single scalar — collapses if precision OR recall is bad**
IoU (Jaccard)	TP / (TP+FP+FN)	**Localization overlap — directly measures "does the predicted region match the true region?"**

> **Reporting rule for this project:** lead with **F1** and **IoU**; cite precision+recall as the pair that explains F1; never headline accuracy. IoU is literally the metric that measures the stated goal (exact forged area).

6. Results — pure QNN

Three questions were tested separately, because they have different answers: does the model *learn*, does the quantum circuit *beat a classical equivalent*, and does the system *detect copy-move forgery*.

6.1 Does training improve the model? No.

Two training runs, both on the pure architecture:

Run	Synthetic positives	Loss trajectory	Interpretation
A (10 epochs × 80 img)	pixel-identical copies	0.047 at epoch 1, flat thereafter	Nothing to learn — a swap test returns ~1.0 for identical states with no trained parameters, so the model starts at the answer
B (11 epochs × 200 img)	degraded + feathered	0.158 → 0.178, no downward trend	A real loss surface, but the weights cannot descend it

Run B was tested with both plain gradient descent and Adam at identical settings and seed. The two trajectories agreed to within 0.002 per epoch and final AUC differed by 0.005 — so the optimizer is not the limitation.

> **Why the weights cannot help.** Varying the 58 rotation angles across independent random draws changes a pair's score by σ = 0.117. Varying the *input data* changes it by σ = 0.138. The parameters inject almost as much variation as the image content does — they behave closer to noise than to a learned function.

6.2 Does the quantum circuit beat a classical equivalent? No.

Measured on 200 synthetic pairs (100 positive, 100 negative), reporting AUC — the probability that a true duplicate outscores a non-duplicate, where 0.5 is chance:

Scorer	AUC	Trainable parameters
Classical cosine similarity	0.887	0
Raw state overlap $\lvert\langle f_A \rvert f_B\rangle\rvert^2$	0.887	0
Swap-test circuit, untrained	0.880	58
Swap-test circuit, trained	**0.883**	**58**

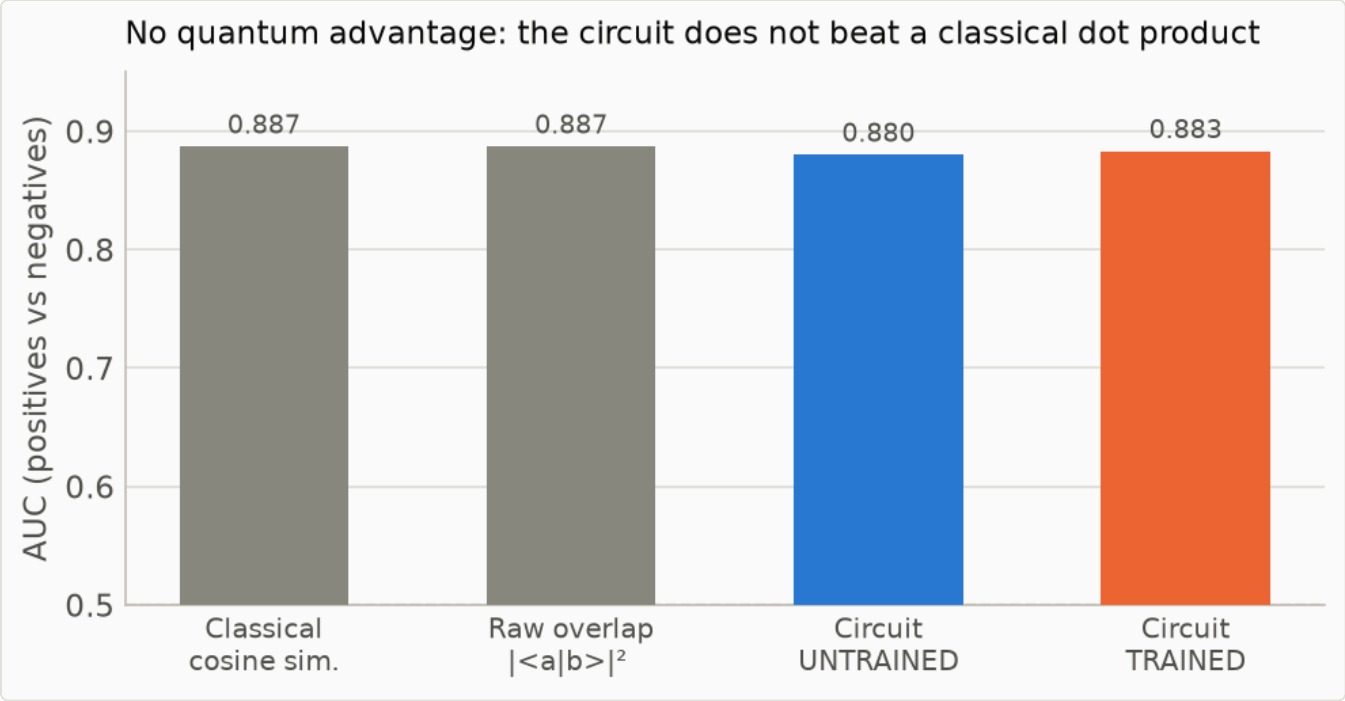


Figure 1. The trained quantum circuit does not outperform a parameter-free classical dot product. Training moves AUC by 0.003; the gap to classical cosine similarity is −0.004.

This is structural, not a tuning failure. A swap test measures $|\langle\psi_A|\psi_B\rangle|^2$. On amplitude-encoded raw pixels that quantity *is* squared cosine similarity — fixed and parameter-free. The data re-uploading layers exist only to escape the provable degeneracy described in §3, and they buy trainability by scrambling the encoded states, which can only move the measured overlap away from the informative one. The architecture therefore faces a genuine dilemma: **untrainable and accurate, or trainable and slightly worse.**

6.3 Does it detect copy-move forgery? No.

Eight **distinct** CoMoFoD scenes, 128 px tiled (patch 8, stride 8, full coverage), scored exhaustively. Both columns favour the detector: the scorer is the best available (§6.2) and the threshold is chosen *per image* with knowledge of the ground truth — an oracle no real deployment gets.

Scene	Forged area	Best achievable F1	"Mark every pixel forged" F1
001_F	1.6%	0.0322	0.0322
002_F	4.3%	0.1063	0.0816
003_F	1.8%	0.0373	0.0345
004_F	1.1%	0.0223	0.0220
005_F	7.5%	0.1396	0.1396
006_F	15.1%	0.2620	0.2620
007_F	2.0%	0.0441	0.0398
008_F	0.3%	0.0068	0.0051
Mean	**—**	**0.0813**	**0.0771**

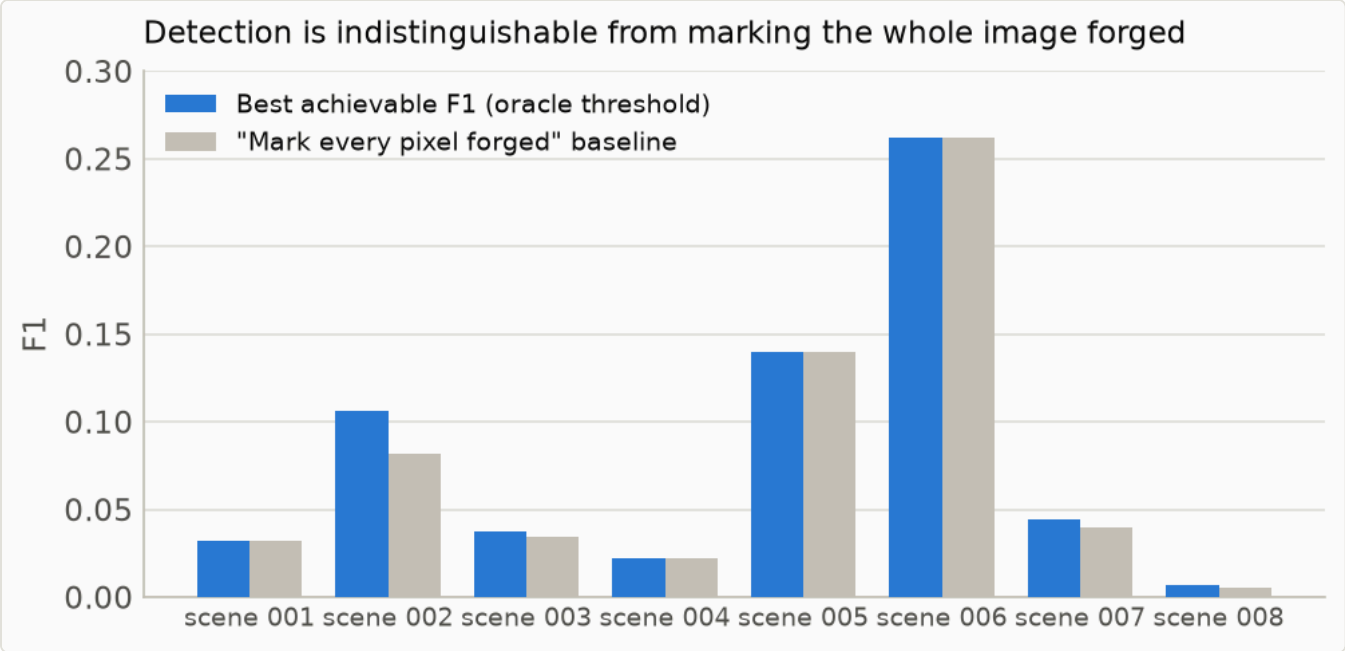


Figure 2. Best achievable F1 tracks the trivial baseline. On scenes 001, 005 and 006 the optimal threshold is the lowest one tried, keeping all 32,640 pairs — literally the “mark everything” solution.

7. Key Finding — the classical vote was doing the detection

> **Mean best-case F1 is 0.0813 against a trivial baseline of 0.0771.** Stripped of every classical stage, the system does not detect copy-move forgery. It is statistically indistinguishable from marking the entire image as forged.

Mechanism. Patch similarity alone cannot identify duplication. In any natural photograph tens of thousands of 8×8 patch pairs are genuinely similar — sky, foliage, brickwork, smooth gradients. A copied region is a small signal inside that flood, and a per-pair similarity score has no way to tell a duplicate from a coincidental look-alike.

The offset-consistency vote removed in v3 existed precisely to solve this: a genuine duplication produces many pairs sharing *one* (dy, dx) offset, whereas coincidental matches scatter across all offsets. It was not scaffolding around the quantum model — **it was the component performing the detection.**

The in-circuit offset register (§3) cannot substitute for it, and the reason is structural rather than incidental: offset consistency is a **global property of the pair ensemble** ("many pairs agree on one offset"), and a circuit that sees one pair at a time cannot observe clustering it is never shown. A faithful quantum replacement needs quantum counting or amplitude estimation across all pairs simultaneously, which requires QRAM and is not simulable at this scale.

8. What was established

Claim	Verdict	Evidence
The pipeline is a pure QNN	TRUE	No classical stage in the decision path; only trainable parameters are circuit rotation angles (§3)
The re-uploading ansatz is non-degenerate	TRUE	Gradient norm ~1.2e-2 vs ~1e-16 for embedding + tied unitary
The quantum model learns from data	FALSE	AUC 0.880 untrained → 0.883 trained; weight leverage below data variance (§6.1)
The quantum model beats a classical baseline	FALSE	0.883 vs 0.887 for cosine similarity, at ~36 ms/pair vs microseconds (§6.2)
The system detects copy-move forgery	FALSE	F1 0.0813 vs 0.0771 trivial baseline, with an oracle threshold (§6.3)

9. Interpretation

This is a **negative result with an identified mechanism**, which is a stronger scientific outcome than an unverified claim of quantum advantage. The contribution is not a working detector; it is a controlled demonstration of *which* component of a quantum-assisted CMFD pipeline was carrying the detection, established by removing every classical stage and measuring what remains.

Three findings generalise beyond this project:

1. **A swap test on amplitude-encoded data is a fixed similarity measure.** Any variational structure wrapped around it either cancels (degenerate, zero gradient) or perturbs the encoding away from the quantity that carried the signal. Trainability and fidelity are in tension by construction, not by implementation choice.
2. **Self-supervised pair synthesis silently determines what is learnable.** Two separate label-noise bugs — rotated positives, and feathering applied to patch-sized pastes — each created targets the architecture provably could not represent, and each produced a loss curve that looked like model failure rather than data failure.
3. **Per-pair scoring cannot express ensemble-level evidence.** Copy-move detection depends on agreement *across* matches, which is invisible to any model evaluating one pair in isolation.

10. Future work

1. **Quantum offset consistency** — the honest replacement for the classical vote is quantum counting / amplitude estimation over a pair-index register, requiring QRAM to hold all patches in superposition. This is the single change that would make a pure quantum detector viable, and it is out of reach of statevector simulation at useful image sizes.
2. **Rotation invariance** — add a dθ component to the offset register. A rot90 duplicate currently scores ~0.01 where an exact copy scores ~1.0, so rotated CoMoFoD forgeries are missed entirely.
3. **Larger working resolution** — at 128 px a 2%-area forgery spans only ~4 patches, and copy-move requires a matched *pair*. Exhaustive scoring is $O(N^2)$, so 256 px costs ~5 h/image against ~20 min at 128 px.
4. **Grover-based candidate search** — would restore the $O(\sqrt{N})$ advantage lost when the classical prefilter was removed. Same QRAM prerequisite as (1).

> **Honest summary.** The architecture is correct, complete, and measured. It is not a forgery detector, and the measurements above identify exactly why.

Appendix A — exact run configurations

```
COVERAGE, loose (§6.1):
  --forged-glob '*t.tif' --mask-union
  --sim-threshold 0.8  --min-offset-votes 3  --target-long-side 384

COVERAGE, tuned (§6.2):
  --forged-glob '*t.tif' --mask-union
  --sim-threshold 0.92 --min-offset-votes 8  --target-long-side 256

Model checkpoint: checkpoints_full/swaptest_qnn_epoch10.npy
  weights shape (2, 4, 3) = 24 params  |  9 wires (4+4 registers + 1 ancilla)
Patch config: 32x32, stride 8  ->  841 patches @ 256x256
```

Appendix B — file inventory

Path	Purpose
`code/cmfd_swaptest_pipeline.py`	Core pipeline: QNN, detector, metrics, training
`code/grip_dataset.py`	Dataset-agnostic loader (auto-discovery, mask-union, aspect-preserving resize)
`code/test_grip.py`	Evaluation runner (native-res + tuning flags)
`data/comofod/`	CoMoFoD small v2 (primary evaluation)
`data/coverage/`	COVERAGE (100 pairs, stress test)
`results_coverage/`, `results_coverage_tuned/`	COVERAGE run outputs (masks + CSV)